October 23, 2026

ECEN 474/704
Lab 8: Prelab
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1. **Three-current-mirror OTA design (Fig. 8-4).** The symmetrical OTA is required to meet Table 8-1 of the lab handout:

Parameter	Requirement
Transconductance, G_m	$> 700\,\mu$S
Slew rate	$> 8\,\text{V}/\mu\text{s}$
Peak-to-peak output swing	$> 1.2\,$V
Load capacitance	$20\,$pF
Power, excluding the bias-reference branch	$< 500\,\mu$W
Power supplies	$V_{DD} = 0.9\,\text{V}, V_{SS} = -0.9\,\text{V}$

Solution.

Device parameters and model assumptions. The device parameters are taken from the single-finger $W_{\text{total}}/L = 5/0.27\,\mu$m rows of Table I in the Lab 1 report:

Parameter	NFET	PFET
$K_P = \mu C_{ox}$ ($\mu\text{A/V}^2$)	215.6	47.57
$\lvert V_{TH}\rvert$ (V)	0.4330	0.4068
λ (V^{-1})	0.1624	0.1945
θ (V^{-1})	0.9999	0.6558

All devices have $L = 0.270\,\mu$m in the initial design. The DC dimensions follow the square-law current model, with body effect, mobility degradation, and DC channel-length modulation neglected. The parameter λ enters only through the output-resistance estimate $r_o \approx 1/(\lambda I_D)$. The measured parameters depend on geometry and bias, so the resulting dimensions define the initial analytical design.

All transistor widths below are *total effective active widths*. In the Lab 1 table, W_{total} already includes all fingers. For equal parallel copies,

$$W_{\text{eff}} = m\, n_f\, w_{\text{finger}}.$$

The total width therefore already accounts for the finger count. The same relation determines the physical finger and multiplicity decomposition under the PDK width convention.

(a) Mirror ratio, current budget, and slew rate. The PMOS mirror ratio and tail current are defined as

$$N = \frac{(W/L)_{5,6}}{(W/L)_{3,4}}, \qquad I_T = I_{\text{tail}}.$$

The handout gives

$$\text{SR} \approx \frac{NI_T}{C_L}, \qquad I_{DD,\text{OTA}} \approx (N+1)I_T.$$

The bias-reference branch is excluded from the latter expression, as in the handout. The slew-rate and power limits require

$$NI_T > (20\times10^{-12})(8\times10^{6}) = 160\,\mu\text{A},$$
$$(N+1)I_T < \frac{500\,\mu\text{W}}{1.8\,\text{V}} = 277.778\,\mu\text{A}.$$

Hence the feasible current interval is

$$\frac{160}{N}\,\mu\text{A} < I_T < \frac{277.778}{N+1}\,\mu\text{A}.$$

It exists only when

$$160(N+1) < 277.778N \quad\Longrightarrow\quad N > 1.3585.$$

A 1:1 PMOS mirror cannot meet both constraints in this model. For $\boxed{N = 3}$, the allowed tail-current interval is $53.333\,\mu\text{A} < I_T < 69.444\,\mu\text{A}$. The selected current is

$$\boxed{I_T = I_{\text{bias}} = 60\,\mu\text{A}}$$

with a 1:1 M9–M10 bias mirror. Then

$$\text{SR} \approx \frac{3(60\,\mu\text{A})}{20\,\text{pF}} = \boxed{9\,\text{V}/\mu\text{s}},$$
$$P_{\text{OTA}} \approx (1.8\,\text{V})(4)(60\,\mu\text{A}) = \boxed{432\,\mu\text{W}}.$$

The ideal current model predicts constant OTA power within the saturation-valid input range. Finite output conductance introduces current variation with common-mode voltage; the DC sweep determines the maximum power over this range. Including the separate $60\,\mu\text{A}$ bias-reference branch gives approximately $540\,\mu\text{W}$. Table 8-1 applies to the OTA power with this branch excluded.

(b) Input transconductance and overdrive. At balanced input, $I_{D1} = I_{D2} = I_T/2 = 30\,\mu\text{A}$ and $G_m = Ng_{m1}$. The minimum input-device transconductance is

$$g_{m1} > \frac{700\,\mu\text{S}}{3} = 233.333\,\mu\text{S}.$$

With $V_{OV1,2} = 0.200\,\text{V}$, the transconductances are

$$g_{m1} = \frac{2(30\,\mu\text{A})}{0.200\,\text{V}} = 300\,\mu\text{S}, \qquad \boxed{G_m = 900\,\mu\text{S} > 700\,\mu\text{S}.}$$

(c) Transistor dimensions and matching. The selected overdrives are $V_{OV} = 0.200\,\mathrm{V}$ for M1–M8 and $V_{OV} = 0.150\,\mathrm{V}$ for the bias/tail devices M9 and M10. At the balanced operating point,

$$\begin{aligned} I_{D1,2,3,4} &= I_T/2 = 30\,\mu\mathrm{A}, \\ I_{D5,6,7,8} &= NI_T/2 = 90\,\mu\mathrm{A}, \\ I_{D9,10} &= 60\,\mu\mathrm{A}. \end{aligned}$$

The square-law current relation gives

$$I_D \approx \frac{1}{2}K_P\frac{W}{L}V_{OV}^2, \qquad \frac{W}{L} = \frac{2I_D}{K_P V_{OV}^2}, \qquad W = L\frac{W}{L}.$$

Using currents in μA and K_P in $\mu\mathrm{A/V}^2$,

$$\begin{aligned} (W/L)_{1,2} &= \frac{2(30)}{215.6(0.200)^2} = 6.9573, \\ (W/L)_{3,4} &= \frac{2(30)}{47.57(0.200)^2} = 31.5325, \\ (W/L)_{5,6} &= \frac{2(90)}{47.57(0.200)^2} = 94.5974, \\ (W/L)_{7,8} &= \frac{2(90)}{215.6(0.200)^2} = 20.8720, \\ (W/L)_{9,10} &= \frac{2(60)}{215.6(0.150)^2} = 24.7372. \end{aligned}$$

Devices	Type	I_D (μA)	V_{OV} (V)	W (μm)	L (μm)
M1, M2	NFET	30	0.200	1.8785	0.270
M3, M4	PFET	30	0.200	8.5138	0.270
M5, M6	PFET	90	0.200	25.5413	0.270
M7, M8	NFET	90	0.200	5.6354	0.270
M9, M10	NFET	60	0.150	6.6790	0.270

The displayed widths are analytical values before grid rounding. The equal-length PMOS dimensions give

$$\frac{W_{5,6}}{W_{3,4}} = 3.$$

A physical implementation with identical unit devices preserves this ratio. The matched pairs are M1/M2, M3/M4, M5/M6, M7/M8, and M9/M10. The M3–M5 and M4–M6 mirrors use consistent unit geometry with the intended 1:3 ratio. NMOS bodies connect to V_{SS}; PMOS bodies connect to V_{DD}.

Dummy MOS devices do not contribute to the active widths above. NFET dummy terminals connect to V_{SS} and PFET dummy terminals to V_{DD}. Consistent dummy multiplicity in the schematic and layout allows LVS to compare the same devices under the PDK convention.

(d) DC bias and input common-mode range. The quiescent input is $V_{CM,Q} = 0.300\,\mathrm{V}$ with zero differential input. The approximate first-stage and mirror-reference voltages are

$$\begin{aligned} V_{S1} = V_{S2} &= 0.300 - (0.4330 + 0.200) = -0.3330\,\mathrm{V}, \\ V_{A1} = V_{A2} &= 0.900 - (0.4068 + 0.200) = 0.2932\,\mathrm{V}, \\ V_B &= -0.900 + 0.4330 + 0.200 = -0.2670\,\mathrm{V}, \\ V_{G9} = V_{G10} &= -0.900 + 0.4330 + 0.150 = -0.3170\,\mathrm{V}. \end{aligned}$$

The reference $V_{o,Q} \approx V_B$ gives equal drain voltages within the matched M5/M6 and M7/M8 pairs. Finite output conductance determines the actual open-loop output bias; the DC model without channel-length modulation does not uniquely fix it. Setting $V_{o,Q} = 0$ therefore requires a suitable differential input offset or feedback arrangement.

At the reference point $V_{o,Q} = -0.2670\,\mathrm{V}$, the saturation checks are

Devices	V_{DS} or V_{SD} (V)	V_{OV} (V)	Margin (V)
M1, M2	0.6262	0.200	0.4262
M3, M4	0.6068	0.200	0.4068
M5, M6	1.1670	0.200	0.9670
M7, M8	0.6330	0.200	0.4330
M9	0.5830	0.150	0.4330
M10	0.5670	0.150	0.4170

All active devices are saturated in the assumed model. The margin is $V_{DS} - V_{OV}$ for an NFET or $V_{SD} - V_{OV}$ for a PFET.

Keeping the input devices and tail source saturated gives

$$\begin{aligned} V_{CM,\mathrm{min}} &\approx V_{SS} + V_{OV10} + V_{TN1} + V_{OV1} \\ &= -0.900 + 0.150 + 0.4330 + 0.200 = -0.1170\,\mathrm{V}, \\ V_{CM,\mathrm{max}} &\approx V_{DD} - |V_{TP3}| - V_{OV3} + V_{TN1} \\ &= 0.900 - 0.4068 - 0.200 + 0.4330 = 0.7262\,\mathrm{V}. \end{aligned}$$

Thus the estimated interval is $-0.1170 < V_{CM} < 0.7262\,\mathrm{V}$, with width $0.8432\,\mathrm{V}$. These bounds neglect body effect. The transistor operating regions in the common-mode DC sweep define the actual input range.

(e) Saturation-based output swing. M6 and M8 set the output-node compliance:

$$V_{o,\max} \approx V_{DD} - V_{OV6} = 0.900 - 0.200 = 0.700\,\mathrm{V},$$
$$V_{o,\min} \approx V_{SS} + V_{OV8} = -0.900 + 0.200 = -0.700\,\mathrm{V}.$$

The estimated saturation-valid interval is therefore

$$\boxed{\Delta V_o = V_{o,\max} - V_{o,\min} = 1.400\,\mathrm{V} > 1.2\,\mathrm{V}.}$$

In the handout's DC-sweep method, the endpoints occur where either output device leaves saturation, and their difference gives the output-swing interval. The maximum symmetric waveform also depends on the quiescent output voltage. For example, about $V_{o,Q} = -0.2670\,\mathrm{V}$,

$$v_{o,\mathrm{pk}} < \min(0.4330, 0.9670)\,\mathrm{V} = 0.4330\,\mathrm{V}.$$

A waveform centered near zero therefore requires bias or feedback that places the quiescent output near zero.

(f) Output resistance, gain, and frequency estimates. Using the Lab 1 λ values at the initial length,

$$r_{o6} \approx \frac{1}{0.1945(90\times10^{-6})} = 57.127\,\mathrm{k\Omega},$$
$$r_{o8} \approx \frac{1}{0.1624(90\times10^{-6})} = 68.418\,\mathrm{k\Omega},$$
$$R_{\mathrm{out}} \approx r_{o6} \parallel r_{o8} = 31.132\,\mathrm{k\Omega},$$
$$A_{v0} \approx G_m R_{\mathrm{out}} = 28.019\ \mathrm{V/V} \quad (28.95\,\mathrm{dB}).$$

With the 20 pF load assumed to dominate output capacitance,

$$f_{p1} \approx \frac{1}{2\pi R_{\mathrm{out}} C_L} = 255.61\,\mathrm{kHz},$$
$$\mathrm{GBW} \approx A_{v0} f_{p1} = \frac{G_m}{2\pi C_L} = 7.162\,\mathrm{MHz}.$$

The factor 2π converts the handout's angular frequency to hertz. The output-pole approximation omits internal-node poles, which also affect the unity-gain frequency and phase margin. The unity-feedback `iprobe`/`stb` configuration determines the phase margin of the transistor circuit. Table 8-1 specifies no numerical gain, GBW, or phase-margin limit.

(g) Design summary.

Quantity	Requirement	First-order estimate
Transconductance	$> 700\,\mu$S	$900\,\mu$S
Slew rate	$> 8\,\mathrm{V}/\mu\mathrm{s}$	$9\,\mathrm{V}/\mu\mathrm{s}$
Saturation-based output interval	$> 1.2\,$V	$1.4\,$V
OTA power, excluding bias branch	$< 500\,\mu$W	$432\,\mu$W
Load capacitance	$20\,$pF	$20\,$pF
Supply rails	$\pm 0.9\,$V	$\pm 0.9\,$V
Low-frequency voltage gain	Characterize	28.95 dB
Dominant pole	Characterize	255.61 kHz
Gain-bandwidth product	Characterize	7.162 MHz
Phase margin	Characterize	Requires `stb`

The calculated transconductance, slew rate, output interval, and OTA power satisfy the specified limits within the stated model. Power follows the ideal current budget, while slew rate assumes fully steered current. Mirror dynamics and parasitic capacitances affect the actual rising and falling slopes.

The DC operating points establish the saturation-valid input range. The maximum OTA power is evaluated over this range, excluding the reference branch specified by the handout. The output-device region transitions in a differential DC sweep determine the output-swing limits.

For AC characterization, the measured transconductance is the ratio of the output current flowing into C_L to the differential input voltage. The first-order relation is

$$\frac{I_{C_L}(s)}{V_{id}(s)} \approx G_m \frac{sR_{\mathrm{out}}C_L}{1 + sR_{\mathrm{out}}C_L}.$$

This current ratio tends to zero at DC, while internal poles produce the high-frequency rolloff. The $700\,\mu$S specification therefore applies to the passband maximum. An input-offset correction establishes an active-region output bias for the open-loop measurement. Rising and falling slew rates and the unity-buffer phase margin complete the dynamic characterization; parasitic extraction includes the effect of the physical layout.

The Lab 6 report provides a comparison between a first-order calculation and transistor-level simulation: the simple amplifier's pre-layout gain is 24.0950 dB, compared with the calculated 27.01 dB. The OTA values above are analytical predictions under the stated device-model assumptions.

Sources.

[1] S. Palermo, *ECEN 474/704 Lab 8: Operational Transconductance Amplifiers*, Fig. 8-4, Table 8-1, and the output-swing and stability-analysis instructions.

[2] Anonymous contributor, *ECEN 704 Lab 1 Report*, Table I: single-finger $W_{\mathrm{total}}/L = 5/0.27\,\mu$m parameter rows.

[3] Anonymous contributor, *ECEN 704 Lab 6 Report*, Tables 1–2.